

GED Science Practice Test 13

PART A — QUESTIONS

GED-SCI13-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a short distraction affects reaction time. Group A completes a reaction-time test in a quiet room. Group B completes the same test while a flashing light appears beside the screen. Both groups use the same computer program and receive the same instructions.

Group | Average reaction time (milliseconds)

A | 245

B | 318

Which conclusion is best supported by the data?

A) The flashing light was associated with slower reaction times.

B) The flashing light was associated with faster reaction times.

C) Both groups had identical reaction times.

D) Reaction time cannot be measured with a computer test.

GED-SCI13-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the power output of four small motors. Power can be calculated as work divided by time.

Motor | Work done (J) | Time (s)

A | 240 | 8

B | 180 | 5

C | 300 | 10

D | 160 | 4

Which motor has the greatest power output?

A) Motor A

B) Motor B

C) Motor C

D) Motor D

GED-SCI13-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist studies sandstone in a desert region. The rock contains rounded sand grains cemented together. The rounded grains suggest the particles were transported before being deposited and later hardened into rock.

Which process most directly formed the sandstone?

- A) Compaction and cementation of deposited sand grains
- B) Freezing of liquid metal in Earth's core
- C) Melting of rock into magma
- D) Fusion of hydrogen inside a star

GED-SCI13-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The adrenal glands release hormones that help the body respond to stress. One hormone, adrenaline, can increase heart rate, widen airways, and help move stored glucose into the blood so muscles have more available fuel.

Which statement best describes the role of adrenaline in this passage?

- A) It helps prepare the body for rapid action during stress.
- B) It breaks rocks into smaller sediment particles.
- C) It stores all inherited information in the stomach.
- D) It prevents blood from carrying oxygen.

GED-SCI13-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A machine does 540 J of work in 18 seconds.

Power = work ÷ time

What is the power of the machine in watts?

Type your answer as a whole number.

GED-SCI13-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how blade shape affects the energy produced by a model water turbine. The student tests curved, flat, and angled blades. The water flow rate and turbine size stay the same in every trial.

Which TWO factors should be kept constant?

- A) Water flow rate
- B) Turbine size
- C) Blade shape
- D) Energy produced
- E) Final measurement

GED-SCI13-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

During meiosis, paired chromosomes can exchange matching sections of DNA. This process is called [D1] and can increase [D2] among offspring.

Dropdowns:

D1) crossing over | condensation | sedimentation

D2) genetic variation | surface temperature | mineral hardness

GED-SCI13-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student holds a metal spoon and a plastic spoon in the same warm room. Both spoons are at room temperature, but the metal spoon feels cooler because it transfers thermal energy away from the hand faster.

Which property explains this difference?

- A) Thermal conductivity
- B) Genetic inheritance
- C) Precipitation rate
- D) Chromosome number

GED-SCI13-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A community wants to reduce soil loss on steep farmland without relying on concrete structures. One plan is to plant strips of deep-rooted vegetation along the slope. The roots can hold soil particles together, and the plant stems can slow moving surface water.

Which result would best support the plan?

- A) Planted slopes lose less soil during storms than unplanted slopes.
- B) Planted slopes produce more earthquakes.
- C) Vegetation stops sunlight from reaching Earth.
- D) Plant roots turn soil into solid metal.

GED-SCI13-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A cardiologist records stroke volume, the amount of blood pumped by the heart with each beat, before and after an endurance training program.

Group | Average stroke volume before (mL/beat) | Average stroke volume after (mL/beat)

Training group | 72 | 88

No-training group | 73 | 74

Which claim is best supported by the data?

- A) The training group showed a greater increase in average stroke volume.
- B) The no-training group showed a greater increase in average stroke volume.
- C) Both groups decreased by the same amount.
- D) Stroke volume was identical after the program.

GED-SCI13-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Shape of the pulley arrangement
- B) Input force needed to lift the load
- C) Same load lifted in each trial
- D) Color of the pulley frame

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI13-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The spleen helps filter blood and supports immune defense. It removes some damaged red blood cells and helps white blood cells respond to certain pathogens in the blood.

Which statement best describes one function of the spleen?

- A) It helps filter blood and supports immune responses.
- B) It produces magma under oceanic crust.
- C) It bends light through a lens.
- D) It stores all body calcium in the lungs.

GED-SCI13-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A soccer ball is kicked across grass. The ball slows because grass exerts a friction force opposite the direction of motion. If the same ball is kicked with the same speed across smooth indoor flooring, it rolls farther.

Which statement best explains why the ball rolls farther indoors?

- A) The smoother floor produces less friction than grass.
- B) The indoor floor removes gravity from the ball.
- C) Grass gives the ball unlimited kinetic energy.
- D) Friction always makes objects speed up.

GED-SCI13-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A forest restoration project plants 2,400 native trees across 8 equal plots.

What is the average number of trees planted per plot?

Type your answer as a whole number.

GED-SCI13-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a new stretching routine improves shoulder range of motion. One group follows the routine for six weeks. A comparison group continues normal activity. Both groups take the same range-of-motion test before and after the study.

Why is the comparison group included?

- A) To help determine whether changes are related to the stretching routine
- B) To guarantee that the stretching group improves
- C) To prevent the researcher from measuring motion
- D) To make shoulder joints stop moving

GED-SCI13-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how launch angle affects the horizontal distance traveled by a foam projectile. The same launcher, same projectile, and same pullback distance are used for every trial.

What is the independent variable?

- A) Launch angle
- B) Horizontal distance traveled
- C) Same projectile
- D) Same pullback distance

GED-SCI13-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A moving object has kinetic energy. If the object's speed increases while its mass stays the same, its kinetic energy [D1]. If the object stops moving, its kinetic energy becomes [D2].

Dropdowns:

D1) increases | becomes inherited | turns into a mineral

D2) zero | maximum | a chromosome

GED-SCI13-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A neurologist measures reflex time in four patients. A shorter reflex time means a faster response.

Patient | Reflex time (milliseconds)

A | 42

B | 58

C | 35

D | 49

Which patient had the fastest reflex response?

- A) Patient A
- B) Patient B
- C) Patient C
- D) Patient D

GED-SCI13-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures how far a spring launcher sends the same ball when different compression distances are used.

Spring compression (cm): 2, 4, 6, 8
Distance traveled by ball (m): 0.7, 1.5, 2.4, 3.6

Which statement is best supported by the data?

- A) Greater spring compression was associated with greater travel distance.
- B) Greater spring compression was associated with shorter travel distance.
- C) Compression distance had no relationship to travel distance.
- D) The ball traveled the same distance every time.

GED-SCI13-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A region tracks two indicators related to drought conditions.

Year	Average soil moisture (%)	Crop irrigation demand (million L)
1	28	410
2	24	485
3	20	560
4	18	625

Which TWO statements are best supported by the data?

- A) Average soil moisture decreased from Year 1 to Year 4.
- B) Crop irrigation demand increased from Year 1 to Year 4.
- C) Soil moisture increased every year.
- D) Irrigation demand was lowest in Year 4.
- E) Both measurements stayed exactly the same.

GED-SCI13-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A genetic counselor explains that a person can inherit a changed copy of a gene that increases risk for a disease. However, lifestyle, environment, and other genes may also affect whether the disease develops.

Which statement is best supported?

- A) Some inherited gene variants increase risk without guaranteeing disease.
- B) A risk-related gene variant always causes disease in every person.
- C) Environmental factors can never affect health.
- D) Genes are not inherited from parents.

GED-SCI13-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student compares two chemical processes. In Process A, the container becomes warmer. In Process B, the container becomes cooler. The student knows that some reactions release thermal energy, while others absorb thermal energy.

Which statement correctly classifies the processes?

- A) Process A is exothermic, and Process B is endothermic.
- B) Process A is endothermic, and Process B is exothermic.
- C) Both processes must be nuclear reactions.
- D) Neither process involves energy transfer.

GED-SCI13-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A text model describes a comet. A comet is made mostly of ice, dust, and rocky material. When it travels near the Sun, some ice changes directly into gas, carrying dust outward and forming a visible tail that points generally away from the Sun.

What causes the visible tail of a comet near the Sun?

- A) Ice changing to gas and carrying dust away from the comet
- B) The comet becoming a planet
- C) The comet stopping its orbit around the Sun
- D) The comet producing its own nuclear light

GED-SCI13-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A genetics clinic reviews 500 patient records. Of these records, 65 include a documented inherited risk variant.

What percent of the records include an inherited risk variant?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI13-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how different protective padding materials affect impact force. The same mass is dropped from the same height onto foam, rubber, cloth, and cardboard padding. A force sensor records the peak impact force.

Which factor is the dependent variable?

- A) Peak impact force recorded by the sensor
- B) Type of padding material
- C) Drop height
- D) Dropped mass

GED-SCI13-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The nucleus of a eukaryotic cell contains chromosomes made of DNA and proteins. DNA carries instructions used to make molecules that affect cell structure and function.

Which statement best describes the nucleus?

- A) It contains genetic instructions in eukaryotic cells.
- B) It pumps blood through the body.
- C) It filters minerals from groundwater.
- D) It reflects light from stars.

GED-SCI13-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the force needed to lift the same object with different ramp lengths. The height of the ramp is the same in every trial.

Ramp length (m): 1, 2, 3, 4

Input force needed (N): 120, 70, 52, 42

Which statement is best supported by the data?

- A) Longer ramps required less input force to reach the same height.
- B) Longer ramps required more input force to reach the same height.
- C) Ramp length had no relationship to input force.
- D) The shortest ramp required the least input force.

GED-SCI13-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four stars by surface temperature.

Star | Surface temperature (K)

A | 3,600

B | 5,400

C | 7,800
D | 12,200

Which star has the highest surface temperature?

- A) Star A
- B) Star B
- C) Star C
- D) Star D

GED-SCI13-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor measures blood urea nitrogen, a waste product that can increase when kidney filtration is reduced.

Patient | Blood urea nitrogen (mg/dL)

- A | 11
- B | 16
- C | 29
- D | 13

Which patient has the highest blood urea nitrogen level?

- A) Patient A
- B) Patient B
- C) Patient C
- D) Patient D

GED-SCI13-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student observes a glowing sign filled with neon gas. When electrical energy is supplied, neon atoms emit visible light. The sign converts electrical energy into light energy and some thermal energy.

Which energy conversion is mainly described?

- A) Electrical energy into light energy
- B) Genetic energy into sediment
- C) Gravitational energy into chromosomes
- D) Chemical energy into groundwater

GED-SCI13-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether a new breathing exercise improves lung capacity. Participants are assigned to a breathing-exercise group or a usual-activity group. Lung capacity is measured using the same device after eight weeks.

Which TWO choices would improve the investigation?

- A) Use the same lung-capacity measuring device for all participants.
- B) Include enough participants in both groups.
- C) Record only the highest lung-capacity result.
- D) Change the measuring device for each participant.
- E) Give one group a longer testing time.

GED-SCI13-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the period of a pendulum with different string lengths.

String length (cm): 20, 40, 60, 80

Period (s): 0.9, 1.3, 1.6, 1.8

Which statement is best supported by the data?

- A) Longer string length was associated with a longer period.
- B) Longer string length was associated with a shorter period.
- C) Period stayed exactly the same for all string lengths.
- D) String length had no relationship to period.

GED-SCI13-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

After a volcanic island forms, bare rock is exposed with little or no soil. Over time, lichens and mosses can begin growing on the rock. As they break down rock and add organic matter, soil slowly forms, allowing grasses and shrubs to grow later.

Which ecological process is described?

- A) Primary succession
- B) Blood clotting
- C) Refraction
- D) Combustion

GED-SCI13-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The respiratory muscles include the diaphragm and muscles between the ribs. When the diaphragm contracts, the chest cavity expands and air moves into the lungs. When the diaphragm relaxes, air can move out.

Which statement best describes the diaphragm's role?

- A) It helps change chest volume so air can move into and out of the lungs.
- B) It filters urea from the blood.
- C) It produces antibodies in bone marrow.
- D) It forms sedimentary rock from sand.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI13-001 — Correct: A — Correct Answer: The flashing light was associated with slower reaction times. — Explanation: Group B had the flashing light and a higher average reaction time, which means a slower response.

GED-SCI13-002 — Correct: D — Correct Answer: Motor D — Explanation: $\text{Power} = \text{work} \div \text{time}$. Motor D has $160 \text{ J} \div 4 \text{ s} = 40 \text{ W}$, the greatest value.

GED-SCI13-003 — Correct: A — Correct Answer: Compaction and cementation of deposited sand grains — Explanation: Sandstone forms when deposited sand grains are compacted and cemented together.

GED-SCI13-004 — Correct: A — Correct Answer: It helps prepare the body for rapid action during stress. — Explanation: Adrenaline increases heart rate, airway opening, and glucose availability during stress.

GED-SCI13-005 — Correct: 30 — Tolerance: exact — Correct Answer: 30 — Explanation: $\text{Power} = 540 \text{ J} \div 18 \text{ s} = 30 \text{ W}$.

GED-SCI13-006 — Correct: A,B — Correct Answer: Water flow rate | Turbine size — Explanation: These factors should remain constant so blade shape is the main variable being tested.

GED-SCI13-007 — Correct: D1=crossing over; D2=genetic variation — Correct Answer: D1=crossing over; D2=genetic variation — Explanation: Crossing over exchanges DNA between paired chromosomes and increases genetic variation.

GED-SCI13-008 — Correct: A — Correct Answer: Thermal conductivity — Explanation: Metal feels cooler because it conducts thermal energy away from the hand faster than plastic.

GED-SCI13-009 — Correct: A — Correct Answer: Planted slopes lose less soil during storms than unplanted slopes. — Explanation: If roots and stems reduce soil movement, planted slopes should show less soil loss.

GED-SCI13-010 — Correct: A — Correct Answer: The training group showed a greater increase in average stroke volume. — Explanation: The training group increased by 16 mL/beat, while the no-training group increased by 1 mL/beat.

GED-SCI13-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Shape of the pulley arrangement; T2=Input force needed to lift the load; T3=Same load lifted in each trial —

Explanation: The pulley arrangement is changed, the input force is measured, and the load is controlled.

GED-SCI13-012 — Correct: A — Correct Answer: It helps filter blood and supports immune responses. — Explanation: The spleen filters blood and helps white blood cells respond to pathogens.

GED-SCI13-013 — Correct: A — Correct Answer: The smoother floor produces less friction than grass. — Explanation: Less friction allows the ball to roll farther before stopping.

GED-SCI13-014 — Correct: 300 — Tolerance: exact — Correct Answer: 300 — Explanation: $2,400 \text{ trees} \div 8 \text{ plots} = 300 \text{ trees per plot}$.

GED-SCI13-015 — Correct: A — Correct Answer: To help determine whether changes are related to the stretching routine — Explanation: A comparison group helps evaluate whether the routine caused the observed change.

GED-SCI13-016 — Correct: A — Correct Answer: Launch angle — Explanation: The launch angle is the factor intentionally changed by the student.

GED-SCI13-017 — Correct: D1=increases; D2=zero — Correct Answer: D1=increases; D2=zero — Explanation: Kinetic energy increases with speed and becomes zero when an object is not moving.

GED-SCI13-018 — Correct: C — Correct Answer: Patient C — Explanation: The fastest reflex response is the shortest time, 35 milliseconds.

GED-SCI13-019 — Correct: A — Correct Answer: Greater spring compression was associated with greater travel distance. — Explanation: The ball traveled farther as spring compression increased.

GED-SCI13-020 — Correct: A,B — Correct Answer: Average soil moisture decreased from Year 1 to Year 4. | Crop irrigation demand increased from Year 1 to Year 4. — Explanation: Soil moisture fell from 28% to 18%, while irrigation demand rose from 410 to 625 million L.

GED-SCI13-021 — Correct: A — Correct Answer: Some inherited gene variants increase risk without guaranteeing disease. — Explanation: The passage explains that disease risk can be influenced by genes, environment, and other factors.

GED-SCI13-022 — Correct: A — Correct Answer: Process A is exothermic, and Process B is endothermic. — Explanation: A process that warms the container releases thermal energy, while a process that cools the container absorbs thermal energy.

GED-SCI13-023 — Correct: A — Correct Answer: Ice changing to gas and carrying dust away from the comet — Explanation: Near the Sun, comet ice can change directly to gas and carry dust outward, forming a tail.

GED-SCI13-024 — Correct: 13 — Tolerance: exact — Correct Answer: 13 — Explanation: $65 \div 500 \times 100 = 13\%$.

GED-SCI13-025 — Correct: A — Correct Answer: Peak impact force recorded by the sensor — Explanation: The dependent variable is the measured result after changing padding material.

GED-SCI13-026 — Correct: A — Correct Answer: It contains genetic instructions in eukaryotic cells. — Explanation: The nucleus contains chromosomes made of DNA, which carries genetic instructions.

GED-SCI13-027 — Correct: A — Correct Answer: Longer ramps required less input force to reach the same height. — Explanation: The input force decreases from 120 N to 42 N as ramp length increases.

GED-SCI13-028 — Correct: D — Correct Answer: Star D — Explanation: Star D has the highest listed temperature, 12,200 K.

GED-SCI13-029 — Correct: C — Correct Answer: Patient C — Explanation: Patient C has the highest blood urea nitrogen level, 29 mg/dL.

GED-SCI13-030 — Correct: A — Correct Answer: Electrical energy into light energy — Explanation: The neon sign uses electrical energy to produce visible light.

GED-SCI13-031 — Correct: A,B — Correct Answer: Use the same lung-capacity measuring device for all participants. | Include enough participants in both groups. — Explanation: A consistent measuring device and enough participants improve reliability.

GED-SCI13-032 — Correct: A — Correct Answer: Longer string length was associated with a longer period. — Explanation: The period increased from 0.9 s to 1.8 s as string length increased.

GED-SCI13-033 — Correct: A — Correct Answer: Primary succession — Explanation: Primary succession begins on bare rock where little or no soil exists.

GED-SCI13-034 — Correct: A — Correct Answer: It helps change chest volume so air can move into and out of the lungs. — Explanation: Diaphragm contraction and relaxation change chest volume, helping ventilation.